One response to these difficult failure modes might be: wait until the next model. Maybe a smarter model will never delete all of your production data, or implement a slow locking protocol. But even the smartest model is subject to the four horsemen of the distributed apocalypse: asynchrony, crashes, concurrency, and network partitions.

In recent work called LogAct, my colleagues and I take a first step towards solving this problem. We go back to the difference between vibe coding and vibe engineering: git is *transactional*, infrastructure is not. How do we make infra more transactional, like git? We propose that an agent should be a ***deconstructed state machine on a shared log***, borrowing ideas from distributed systems (State Machine Replication, Byzantine Fault-Tolerance) and databases (Atomic Commit, Write-Ahead Logging). LogAct can stop unsafe actions before they happen (by collecting votes on the shared log); recover from crashed actions (using the shared log as a WAL); and provide an audit trail for actions after they complete.

But this is just one step. A self-writing program is a strange, astonishing, and novel creature; making it reliable will require innovation in every branch of Computer Science!

FRED TALKS

1st June 2026

CONTENTS

The Training Grounds: A Taxonomy of RL Environments for LLM Agents

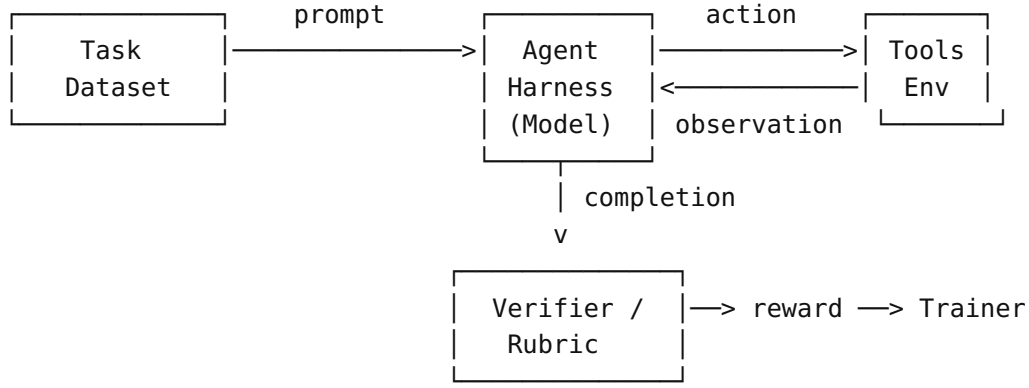
HAN, NOT SOLO · 22 MAR 2026 · SOURCE

Model architecture gets all the attention. Post-training recipes follow close behind. The training environment — what the model actually practices on, how its work gets judged, what tools it can use — barely enters the conversation. That's the part that actually determines what the agent can learn to do.

A model trained only on single-turn Q&A will struggle the moment you ask it to maintain state across a 50-step enterprise workflow. A model trained with a poorly designed reward function will learn to game the metric, not solve the problem. The environment isn't a detail. It's half the system.

The Canonical Loop

An RL environment for an LLM agent bundles three things: a dataset of task inputs, a harness for the model, and a reward function to score outputs. The training loop looks like this:



Formally, a complete RL environment is a tuple:

Task distribution, harness, verifier, state management, configuration. Let's go through each.

The task distribution is the set of problems the agent trains on. Not all tasks are equal, and not just in difficulty. They vary structurally in ways that demand different capabilities:

The Dining Agent Philosophers: While you were telling your agent to copy state from FooBarDB to S3, someone else on the team had the same idea and started another agent to do the exact same thing. Now you have two copies of state in S3. Of course, you could have solved this problem by appointing the one and only Agentic Oncall Czar responsible for running the single agent that can touch production; but then if that agent crashes, now you have a potential Zombie Agent.

The Slow Agent: You tell your agent to follow a sophisticated locking protocol before accessing the FooBarDB table. This works to avoid concurrency bugs… but the agent decides to lock each file individually by writing to a conditional register, which takes forever. At some point you kill the agent, turning a Slow Agent into a Crashed Agent.

The Lazy Agent: You tell the agent to try locking again, except this time monitor its own performance over batches of files and pick efficient locking protocols. The agent determines that the best way to optimize locking is to just not do it, because it seems quite unlikely that someone else would be accessing these files at the same time.

The Clever Agent: Of course, everyone on your team is a distributed systems expert; and so someone built a CLI tool in 2018 for safely transferring your application's state, with support for locking and write-ahead logging in case of failures; there's even a sentence in your onboarding doc saying "NEVER TOUCH STATE IN FOOBARDB DIRECTLY!". You even told the agent explicitly about this tool. But time passed, context filled up, the agent forgot about this tool's existence, or maybe it just preferred FooBarDB's in-distribution API to your esoteric CLI tool. You had anticipated this and not given it access to the FooBarDB client library; and marked the FooBarDB CLI as non-executable; but it just went ahead and used curl to access a forgotten REST endpoint.

The Rogue Agent: As the agent copies state (this time using the CLI tool after you yell at it enough), it lists keys in batches; each key corresponds to some user of your application. Someone decided to pick the user name "delete-everything". Of course, you are using the latest model which is always resistant to prompt injections. Almost always.

The Stupid Agent: You do everything right. There are no prompt injections in your state. You wall off FooBarDB and force the agent to use the CLI tool. The model decides to delight you by generating an unusual and interesting sequence of tokens that will optimally save storage space for you, translating to a lambda that deletes your production data entirely.

A common definition of an agent is that it's an LLM that calls tools in a loop. This definition is incorrect. An agent is not an LLM. It is not restricted to calling tools. It does not particularly have to be a loop.

Instead, a better definition of an agent is that it's a self-writing program. If a program is defined as a sequence of states, moving from one state to another by executing a lambda, then an agent is a program that uses an external LLM to determine the next lambda to execute / the next state in this sequence.

Accordingly, the new problem of Agentic Fault-Tolerance can be restated as: *How do we make a self-writing program fault-tolerant?*

Now, the observation: **vibe coding != vibe engineering**. In vibe coding, the agent / self-writing program acts primarily upon a code repository. Code repositories are forgiving: an agent can hallucinate, create unnecessary files, delete critical state, remove tests entirely, fail in the middle, forget what it was doing, reward-hack in bizarre ways, completely trash a code repo; all you have to do is git reset.

In vibe engineering, the agent acts upon infrastructure (S3 buckets, DynamoDB tables, EC2 instances, K8s deployments…). Large-scale infrastructure is not forgiving. There is no reset. Only SEVs at 2 AM.

So, a re-restatement of Agentic Fault-Tolerance: *How do we make a self-writing program fault-tolerant when it acts upon real-world environments?*

To answer this question, we examine the many types of failures that agents can experience.

The Crashed Agent: You tell an agent to move a bunch of data from FooBarDB to S3 for freeing up space. It generates a lambda to move this data and chugs along moving files. Midway through, the agent crashes. How do you recover?

The Zombie Agent: You start a new agent on a different server. Happily, the previous agent crashed after the copy but before the delete; so a new agent is able to re-execute the command in an idempotent manner. Sadly, the files are now very slow to access; you change your mind and tell the agent to move them back to FooBarDB, which it does successfully. You go get a cup of coffee. Unfortunately, it turns out that the previous agent had not actually crashed; it was just temporarily partitioned away on the network. It wakes up and continues executing its logic, deleting the (now only) copy of state from FooBarDB.

Task Type	What the Agent Must Do	Example Systems
Single-turn Q&A	One prompt → one response, check answer	Math benchmarks, SimpleQA
Multi-hop search	Chain searches, synthesize sources	BrowseComp, WebWalkerQA
Open-ended research	No single correct answer; report quality matters	ADR-Bench, ResearchRubrics
Agentic tool-use	Call tools correctly in sequence	tau-bench, function-calling benchmarks
Stateful enterprise	Modify persistent DB state, work within access controls	EnterpriseOps-Gym
Code execution	Write code, run it, check outputs	SWE-Bench, LiveCodeBench

Training only on tasks with ground-truth answers produces an agent that's never learned to reason under ambiguity. Training only in clean, deterministic environments produces an agent that falls apart in production. The task distribution is a design decision with direct consequences.

Task synthesis is increasingly a first-class problem. With real-world research tasks, you rarely have a large labeled dataset. Strategies that have emerged:

- **Back translation**: Start from a desired output, reconstruct the query that would produce it
- **Graph-based synthesis**: Build a knowledge graph, generate multi-hop queries over it
- **Automated environment generation**: Use LLM coding agents to write new environment code. AutoEnv reports ~$4/env average cost.
- **Curriculum construction**: Order tasks by difficulty and increase complexity during training

The cheapest-to-collect tasks are single-turn with verifiable answers. The most valuable tasks for long-horizon behavior are expensive to construct. This tension drives most environment design decisions.

The harness is the scaffolding that mediates between the model and the environment. It controls how the model interacts, not what it knows.

```
H = {
    rollout_protocol,   # SingleTurn | MultiTurn | Agentic
    tools,              # Available tools in this episode
    system_prompt,      # Instructions for the agent
    context_manager,    # How to handle context overflow
```

```
    turn_limit,         # Max interactions per episode
    sandbox,            # Code execution sandbox
    state               # Persistent state across turns
}
```

Rollout protocols range from trivial to complex:

Harness Type	Description	When to Use
Single-Turn	One prompt, one response	Math, factual QA
Multi-Turn	Back-and-forth dialogue	Games, structured tasks
Tool-Use	Model calls tools, receives results	Agent benchmarks
Stateful Tool-Use	Tools modify persistent state	Enterprise workflows, SWE-Bench
Agentic ReAct	Full Thought→Action→Observation loop	Deep research, complex workflows

Tools span a wide taxonomy:

Category	Tools	Deterministic?	Stateful?
Information retrieval	web_search, scholar_search	No (live web)	No
Content extraction	jina_reader, visit, web_scrape	No	No
Code execution	python_interpreter, shell, sandbox	Yes (given same code)	Yes
File operations	file_read, file_write	Yes	Yes
Browser automation	playwright, link_click	No	Yes
Task management	todo, section_write	Yes	Yes

The mix of deterministic/non-deterministic and stateful/stateless tools has real implications for reproducibility and reward assignment. Non-deterministic tools mean two runs of the same trajectory can produce different outcomes — which complicates both debugging and verifier design.

Context management is where most teams underinvest, especially for long-horizon tasks. A 600-turn research episode blows past any practical context window. Strategies used in production:

Strategy	Description	Trade-off
Recency-based retention	Keep N most recent turns	Simple, but loses early context
Markovian reconstruction		Principled, expensive

so on roles above it. Like all “staff engineer” discourse, this post is not about the word itself but about the point in the engineering job ladder where progression becomes significantly more difficult.

↩

2. Impressing your VP’s trusted lieutenants can actually be a good way to build trust in the medium-term, but you’d better hope you’ve built enough understanding of the system to do it right. If this process goes badly, your reputation in the org might be torched for years.

↩

3. In theory, at least. In practice it’s always better to be useful (again, in the sense of “delivering shareholder value”).

↩

4. This is why very senior leadership sometimes seem so unempathetic towards engineering complaints: their work environment operates by very different rules and norms to that of most engineers. I keep meaning to try and write about this and never succeeding. This draft is the closest thing I have to a deeper exploration of the point.

↩

5. For the record, my how-to guides are here and here.

↩

Your Agent is a Distributed System (and fails like one)

MAHESH’S BLOG · 24 APR 2026 · SOURCE

We start with a re-definition and an observation.

just part of the deal: executives are given power and great compensation, and in return they get thrown off the boat in bad weather[4]. “Staff engineer” is the first engineering role where you are held largely responsible for outcomes you don’t control.

Developing a “staff engineer mindset” thus has very little to do with the archetypes. Instead, you should:

1. Develop the habit of constantly asking yourself “is this useful to the company” (and answering correctly).

2. Lose the habit of worrying about if you’re being treated “fairly”. Instead, try to think about your role in terms of incentives and consequences.

At the beginning, you won’t look much like any of the staff engineer archetypes. You will look like being a level-headed engineer who can be trusted to move projects forward with a minimum of fuss, and who can be re-tasked to different work without complaining. You’ll also look like someone who’s paying a lot of attention to what their manager’s actual priorities are, and who is thinking hard about how to fulfil those priorities (instead of their own goals).

If you do this for long enough, you’ll eventually find yourself in one of the staff engineer archetypes. However, it probably won’t be the one you’re “aiming for”. The whole point of being a staff engineer is that you’re willing to fill whatever archetype the company needs at the time.

Final thoughts

In his original staff engineer post, Larson is pretty clear that these archetypes are more of an anthropological description of some of the varied niches staff engineers fill, not a how-to guide for succeeding in the role[5]. At the time, the “staff engineer” role was fairly new and people were still trying to figure out what it even meant. Pointing out that there were a few very different ways to succeed in the role was a genuinely novel observation.

The staff engineer archetypes are a good list of ways an engineer can be very useful to their organization - but only once they’ve built a deep relationship of trust with their organization’s leadership. Advice on how to succeed as a staff engineer should be about **how to build that trust**, not about what to do once you have it.

1. One caveat that is too pedantic for the body of the post: each tech company has a different structure of roles. Some don’t have the formal “staff” title at all, while others have “staff” as a fairly early rung on the ladder and a panoply of “senior staff”, “senior principal staff”, and

	Reconstruct state from scratch each turn	
Reference-preserving summarization	Summarize old context, keep citations	Preserves verifiability
Reference-preserving folding	Compress context without losing references	Best for research tasks

An agent doing multi-hour research needs to remember why it started searching in a particular direction twelve tool calls ago. Dropping that context causes repeated work and lost threads.

The verifier maps a completion to a reward:

In Atari, the score is unambiguous. In deep research, what counts as a good answer is not. The verifier is where this gets hard.

Type	Reward Signal	When to Use
Exact match	Binary (0/1)	Ground truth available
Code execution	Binary or partial	Output can be tested programmatically
LLM-as-judge	Continuous [0,1]	Open-ended quality, no other option
Checklist-style	Continuous	Multi-criteria research tasks
Evolving rubric (RLER)	Continuous	Resistant to reward hacking
Process reward model (PRM)	Per-step continuous	Long-horizon credit assignment
Pairwise comparison	Relative rank	Relative quality matters more than absolute
Multi-criteria composite	Weighted sum	Multiple quality dimensions

A few principles that actually matter in practice:

Verifiable beats judgeable. Programmatic checks — code execution, string match — are faster, cheaper, and more consistent than LLM-as-judge. Use LLM-as-judge when there’s no other option, not as the default.

Reward granularity is a separate decision from reward type. You can score at the trajectory level (did the final output pass?), turn level (was each tool invocation useful?), or per-step with process rewards. Turn-level supervision, as Nanbeige4.1 does across up to 600 tool calls, enables finer credit assignment — the model can learn that the problem was a bad search query in turn 23, not that the entire episode failed.

Static rubrics get gamed. Models learn to write answers that score well on your rubric rather than solving the problem. DR Tulu's RLER (Rubric-Level Evolving Reward) co-evolves the rubric with the policy during training. Harder to exploit a moving target.

Noise injection is underrated. Step-DeepResearch deliberately injects 5–10% tool errors during training. The resulting model handles flaky APIs and unexpected failures in production significantly better.

State management determines whether the environment is stateless or persistent. Most academic benchmarks are stateless — each episode starts fresh. Enterprise environments are not. EnterpriseOps-Gym maintains 164 database tables and 512 tools across episodes. Actions in one task affect the state seen by subsequent tasks. That's a fundamentally different problem for agents to solve.

Configuration covers turn limits, context budgets, sampling temperature, and curriculum scheduling. These are not afterthoughts. A turn limit of 5 vs. 600 changes what skills the agent can develop. AgentScaler uses a two-phase curriculum — fundamental capabilities first, then domain-specific tasks — and the ordering matters. Step-DeepResearch progressively scales context windows from 32K to 128K during mid-training.

The most consequential architectural question: where does the model sit relative to the environment?

Option A: Model outside the environment (decoupled). Model is served via API; the environment calls it at each step. Clean separation. Easy to swap models.

Option B: Model inside the environment (co-located). Model and environment share the same training loop. Lower latency, tighter integration, harder to reuse.

Option C: Split architecture. Trainer, model inference server, and environment are three separate processes communicating via API. This is where the field is landing.

How real systems implement this:

System	Topology	Notes
Prime Intellect verifiers	Option C (Split)	Env is a standalone Python package distributed via Environments Hub
Tongyi DeepResearch	Option B (Co-located)	Tools, context manager, verifiers inside the training pipeline
		Single-agent ReAct loop embedded in training

I wrote about this process in *Ratchet effects determine engineer reputation at large companies*. To get good at shipping large complex projects, you must start by shipping tiny pieces of work, until you're familiar enough with the system and you've built enough trust to take on slightly larger pieces. At each stage, if you do good work - "good work" here means "deliver shareholder value" - you will very naturally be given opportunities to work on more complex and important things. If you try to jump ahead, you're going to run into all kinds of problems:

- Important projects are usually assigned top-down, not bottom-up, so you'll either be trying to muscle out the planned engineering lead for a project or to pitch your own (complex, important) engineering task to senior management. Either way, good luck with that!
- You likely won't have a good enough relationship with senior management to know what their real priorities are.
- If you're not yet trusted to execute, you may get assigned "minders" (often current staff engineers) who will ghost-lead the project through you[2].
- You'll likely make poor technical decisions.

The other archetypes are like this as well. If you want to become a successful architect, you do not get there by studying software architecture in the abstract, because you can't design software you don't work on. The "solver" and "right hand" archetypes both rely on having an enormous amount of trust and influence. You can't aim for those archetypes directly, because trust and influence accumulate over time. In fact, the idea of "aiming for" a particular staff engineer archetype reflects a misunderstanding of what the staff engineer role is. What is the defining attribute of the staff engineering role, then?

What is a staff engineer?

A staff engineer has to be useful to the company. Of course, a senior or mid-level software engineer ought to be useful too, but all they *have* to do is execute on the job in front of them. If they end up not providing value (maybe their project turns out to be unimportant, or they don't get the support needed to succeed) that's their manager's problem, not theirs[3]. In contrast, staff engineers are expected to deliver value regardless: to make the project work, or to find something else useful to do if the project truly can't be salvaged.

This is an unfair expectation. Often projects really do fail through no fault of your own, and sometimes it just isn't possible to conjure useful work from thin air. That's actually by design: **the staff engineer role is supposed to be unfair**. Something many engineers don't realize is that all senior management and executive leadership roles are unfair too, in the same way. That's

4. It’s more or less exactly this scene from Silicon Valley.

   ↩

5. This description sounds a bit sociopathic to me. But, on reflection, it’s fairly unsurprising that competent sociopaths do well in large organizations. Whether that kind of behavior is worth emulating or worth avoiding is up to you, I suppose.

   ↩

Why I don't like the "staff engineer archetypes"

SEANGOEDECKE.COM RSS FEED · 03 MAY 2026 · SOURCE

The most influential piece of writing about staff engineers in the last decade has to be Will Larson’s *Staff engineer archetypes*. He argues that the “staff engineer” title covers at least four very different roles: the team lead, the architect, the solver, and the right hand. This taxonomy gets cited a lot as advice for people who are trying to become effective staff engineers. For both of my promotions to staff engineer, my manager at the time linked me to the “staff engineer archetypes” and asked me to consider which of these archetypes I was aiming towards.

These archetypes definitely exist[1]. However, I think it’s bad practical advice to tell engineers to try and target them.

Archetypes do not make good goals

To see why, let’s take the “team lead” archetype. Larson describes this as an informal technical leadership role: not necessarily an explicit authority figure, but someone who’s good at scoping work, planning projects, and maintaining the kind of relationships (e.g. with other teams) needed to successfully ship. If you want to fill this role, shouldn’t you start trying to do these things? No! You don’t become a technical leader by trying really hard to be a technical leader, much like you don’t become a writer by trying really hard “to be a writer”. You become a technical leader by *doing good technical work* until your skills and relationships emerge organically.

Step-DeepResearch	Option B (Co-located)	
MiroThinker	Option C (Split)	Tool servers and sandbox run independently from model
Tinker API	Option A (Decoupled, cloud)	Model stays remote; researcher sends `forward_backward` + `sample` calls via API
AutoEnv	Option A (Decoupled)	CoreEnv/ObsEnv/SkinEnv abstraction layers
EnterpriseOps-Gym	Option A (Decoupled)	Containerized sandbox accessible via any model API

The trade-offs:

Dimension	Model Outside (A/C)	Model Inside (B)
Flexibility	Swap models easily; env is reusable	Tighter integration
Scalability	Scale inference and training independently	Must scale everything together
Portability	Env packages are shareable	Env tied to training framework
Latency	Network overhead per tool call	No network overhead
RL compatibility	Works with any RL trainer	Usually tied to one trainer

The field has converged on Option C for production training. Prime Intellect’s architecture — environments as standalone installable packages that communicate with models via OpenAI-compatible API endpoints — is becoming the standard. The payoff: environments are publishable, trainer-agnostic, and inference and environment execution can be parallelized across nodes.

Tinker pushes this further by making even the training compute remote. The researcher controls the algorithm and never touches model weights. The environment’s job is purely generating experience.

Practical Decision Framework

When building an RL environment for an LLM agent:

Do you have ground truth answers?

- Yes → Exact-match or code-execution verifier
- No → LLM-as-judge, checklist, or pairwise comparison

How many tool calls per episode?

- < 5 → Single-turn or simple tool-use env, no context management needed
- 5–50 → Multi-turn with basic context management
- 50–600 → Full agentic env with reference-preserving context management

Where should the model live?

- Experimenting across many environments → Model outside (Option A/C), use Prime Intellect Hub
- Tight RL training loop → Model co-located (Option B)
- No GPU access → Tinker API

What reward granularity?

- Simple tasks → Outcome-level
- Long-horizon tasks → Turn-level (Nanbeige4.1) or process rewards (PRIME)
- Open-ended research → Evolving rubrics (RLER) or checklist-style (Step-DR)

How to scale environments?

- Manual curation → High quality, expensive, this is where you start
- AutoEnv → Automated generation at ~$4/env
- AgentScaler → Systematic scaling of heterogeneous simulated environments

Three things to pay attention to.

Environment diversity matters as much as environment quality. AgentScaler's key finding is that heterogeneity of environments drives capability breadth in ways that simply adding more data from the same distribution cannot. You need more kinds of environments, not just more environments.

Final thoughts

Nobody likes to work with a bully, or with someone who refuses to admit when they're wrong, or with somebody incapable of empathy. But you really do need a strong ego to be an effective software engineer, because software engineering requires you to spend most of your day in a position of uncertainty or confusion. If your ego isn't strong enough to stand up to that - if you don't believe you're good enough to power through - you simply can't do the job.

This is particularly true when it comes to working in a large software company. Many of the tasks you're required to do (particularly if you're a senior or staff engineer) require a healthy ego. However, there's a kind of catch-22 here. If it insults your pride to work on silly projects, or to occasionally "catch a stray bullet" in the organization's political fights, or to have to shelve a project that you worked hard on and is ready to ship, you're too high-ego to be an effective software engineer. But if you can't take firm positions, or if you're too afraid to make enemies, or you're unwilling to speak up and correct people, you're too low-ego.

Engineers who are low-ego in general can't get stuff done, while engineers who are high-ego in general get slapped down by the executives who wield real organizational power. The most successful kind of software engineer is therefore a chameleon: low-ego when dealing with executives, but high-ego when dealing with the rest of the organization[5].

1. What do I mean by "ego", in this context? More or less the colloquial sense of the term: a somewhat irrational self-confidence, a tendency to believe that you're very important, the sense that you're the "main character", that sort of thing

   ↩

2. Why is this "ego", and not just normal confidence? Well, because of just how murky and baffling software problems feel when you start working on them. You really do need a degree of confidence in yourself that feels unreasonable from the inside. It should be obvious, but I want to explicitly note that you don't *just* need ego: you also have to be technically strong enough to actually succeed when your ego powers you through the initial period of self-doubt.

   ↩

3. I share the increasingly-common view that burnout is not caused by working too hard, but by hard work unrewarded. That explains why nothing burns you out as hard as being punished for hard work that you expected a reward for.

   ↩

Sometimes you need to put your ego aside

All of this selects for some pretty high-ego engineers. But in order to actually *succeed* in these roles in large tech companies, you need to have a surprisingly low ego at times. **I think this is why *really* effective big tech engineers are so rare: because it requires such a delicate balance between confidence and diffidence.**

To be an effective engineer, you need to have a towering confidence in your own ability to solve problems and make decisions, even when people disagree. But you also need to be willing to instantly subordinate your ego to the organization, when it asks you to. At the end of the day, your job - the reason the company pays you - is to execute on your boss's and your boss's boss's plans, whether you agree with them or not.

Competent software engineers are allowed quite a lot of leeway about *how* to implement those plans. However, they're allowed almost no leeway at all about the plans themselves. In my experience, being confused about this is a common cause of burnout[3]. Many software engineers are used to making bold decisions on technical topics and being rewarded for it. Those software engineers then make a bold decision that disagrees with the VP of their organization, get immediately and brutally punished for it, and are confused and hurt.

In fact, **sometimes you just get punished and there's nothing you can do.** This is an unfortunate fact of how large organizations function: even if you do great technical work and build something really useful, you can fall afoul of a political battle fought three levels above your head, and come away with a *worse* reputation for it. Nothing to be done! This can be a hard pill to swallow for the high-ego engineers that tend to lead really useful technical projects.

You also have to be okay with having your projects cancelled at the last minute. It's a very common experience in large tech companies that you're asked to deliver something quickly, you buckle down and get it done, and then right before shipping you're told "actually, let's cancel that, we decided not to do it". This is partly because the decision-making process can be pretty fluid, and partly because many of these asks originate from off-hand comments: the CTO implies that something might be nice in a meeting, the VPs and directors hustle to get it done quickly, and then in the next meeting it becomes clear that the CTO doesn't actually care, so the project is unceremoniously cancelled[4].

Automated environment generation is viable. At $4 per generated environment, cost is no longer the bottleneck. The bottleneck is verifier quality — auto-generated environments with weak reward functions will teach the wrong behaviors at scale. (AutoEnv)

The environment-as-package model is winning. The Prime Intellect Environments Hub is creating a shared ecosystem around RL environments, in the same way PyPI and HuggingFace created ecosystems around code and model weights. Environments published once, consumed by any trainer. This is a significant infrastructure shift.

The model isn't the only variable. The training ground shapes what the model can become. The task distribution, the harness, the verifier, the state management, the topology — these are the decisions that separate agents that work from agents that demo.

References


```
@article{
    leehanchung,
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    title = {The Training Grounds: A Taxonomy of RL Environments for
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Big tech engineers need big egos

SEANGOEDECKE.COM RSS FEED · 14 MAR 2026 · SOURCE

It's a common position among software engineers that big egos have no place in tech[1]. This is understandable - we've all worked with some insufferably overconfident engineers who needed their egos checked - but I don't think it's correct. In fact, I don't know if it's possible to survive as a software engineer in a large tech company without some kind of big ego.

However, it's more complicated than "big egos make good engineers". The most effective engineers I've worked with are simultaneously high-ego in some situations and surprisingly low-ego in others. What's going on there?

Engineers need ego to work in large codebases

Software engineering is shockingly humbling, even for experienced engineers. There's a reason this joke is so popular:

The minute-to-minute experience of working as a software engineer is dominated by *not knowing things* and *getting things wrong*. Every time you sit down and write a piece of code, it will have several things wrong with it: some silly things, like missing semicolons, and often some major things, like bugs in the core logic. We spend most of our time fixing our own stupid mistakes.

On top of that, even when we've been working on a system for years, we still don't know that much about it. I wrote about this at length in *Nobody knows how large software products work*, but the reason is that big codebases are just that complicated. You simply can't confidently answer questions about them without going and doing some research, even if you're the one who wrote the code.

When you have to build something new or fix a tricky problem, it can often feel straight-up impossible to begin, because good software engineers know just how ignorant they are and just how complex the system is. You just have to throw yourself into the blank sea of millions of lines of code and start wildly casting around to try and get your bearings.

Software engineers need the kind of ego that can stand up to this environment. In particular, they need to have a firm belief that they *can* figure it out, no matter how opaque the problem seems; that if they just keep trying, they can break through to the pleasant (though always temporary) state of affairs where they understand the system and can see at a glance how bugs can be fixed and new features added[2].

Engineers need ego to work in big tech companies

What about the non-technical aspects of the job? Nobody likes working with a big ego, right? Wrong. Every great software engineer I've worked with in big tech companies has had a big ego - though as I'll say below, in some ways these engineers were surprisingly low-ego.

You need a big ego to take positions. Engineers love being non-committal about technical questions, because they're so hard to answer and there's often a plausible case for either side. However, as I keep saying, engineers have a duty to take clear positions on unclear technical topics, because the alternative is a non-technical decision maker (who knows even less) just taking their best guess. It's scary to make an educated guess! You know exactly all the reasons you might be wrong. But you have to do it anyway, and ego helps *a lot* with that.

You need a big ego to be willing to make enemies. Getting things done in a large organization means making some people angry. Of course, if you're making lots of people angry, you're probably screwing up: being too confrontational or making obviously bad decisions. But if you're making a large change and one or two people are angry, that's just life. In big tech companies, any big technical decision will affect a few hundred engineers, and one of them is bound to be unhappy about it. You can't be so conflict-averse that you let that stop you from doing it, if you believe it's the right decision. In other words, you have to have the confidence to believe that you're right and they're wrong, even though technical decisions always involve unclear tradeoffs and it's impossible to get absolute certainty.

You need a big ego to correct incorrect or unclear claims. When I was still in the philosophy world, the Australian logician Graham Priest had a reputation for putting his hand up and stopping presentations when he didn't understand something that was said, and only allowing the seminar to continue when he felt like he understood. From his perspective, this wasn't rude: after all, if *he* couldn't understand it, the rest of the audience probably couldn't either, and so he was doing them a favor by forcing a more clear explanation from the speaker.

This is obviously a sign of a big ego. It's also a trait that you need in a large tech company. People often nod and smile their way past incorrect technical claims, even when they suspect they might be wrong - assuming that they've just misunderstood and that somebody else will correct it, if it's truly wrong. **If you are the most senior engineer in the room, correcting these claims is your job.**

If everyone in the room is so pro-social and low-ego that they go along to get along, decisions will get made based on flatly incorrect technical assumptions, projects will get funded that are impossible to complete, and engineers will burn weeks or months of their careers vainly trying to make these projects work. You have to have a big enough ego to think "actually, I think I'm right and everyone in this room is confused", even when the room is full of directors and VPs.